Single Oral Practice 1

Module 5 - Single Oral Administration · Daily practice sheet · No date

Name _____ Date _____

Procainamide is used for the treatment of ventricular tachyarrhythmia. It is administered intravenously, intramuscularly, and orally (Canada), and its therapeutic range is 4 to 8 mcg/mL. When a 750 mg dose is administered intravenously to a normal healthy subject: the elimination half-life is 3 hr; the apparent volume of distribution is 140 L or 2 L/kg; 65% is excreted in the urine; 35% is metabolized. When a 250 mg tablet is administered orally to a normal healthy subject: the absorption rate constant is 2.8 hr^{-1}; fraction of dose absorbed is 85.54%. Determine the following from the data provided:

Part 1 Total body clearance (and renal and metabolic clearance). What is the total body clearance?

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Part 2 Time to peak following the oral administration of a 250 mg dose.

Part 2 Maximum plasma concentration following the oral administration of a 250 mg dose.

Part 2 Maximum plasma concentration following the oral administration of a 500 mg dose.

Part 3 Whether the 250 mg tablet will provide a maximum plasma concentration sufficient to control the arrhythmia.

A. Yes: 1.22 mg/L is above 1 mcg/mL
B. No: its peak, 1.22 mg/L, is below the 4 mcg/mL lower limit
C. It cannot be judged without the AUC
D. Yes, after unit conversion 1.22 mg/L is 1220 mcg/mL

Part 4 If the 250 mg tablet is insufficient to control the arrhythmia, how many tablets (250 mg strength) will be required to control the arrhythmia?

A. 2 tablets
B. 4 to 6 tablets
C. 4 to 7 tablets
D. 3 to 6 tablets

Part 5 In a 70 kg patient with renal impairment, the elimination half-life is reported to be 14 hours. Assuming no change in apparent volume of distribution, absorption rate constant, and fraction absorbed systemically, determine the total body, renal, and metabolic clearances following the administration of a 250 tablet to this subject. What is the total body clearance?

Part 5 In a 70 kg patient with renal impairment, the elimination half-life is reported to be 14 hours. Assuming no change in apparent volume of distribution, absorption rate constant, and fraction absorbed systemically, determine the total body, renal, and metabolic clearances following the administration of a 250 tablet to this subject. What is the renal clearance?

Part 5 In a 70 kg patient with renal impairment, the elimination half-life is reported to be 14 hours. Assuming no change in apparent volume of distribution, absorption rate constant, and fraction absorbed systemically, determine the total body, renal, and metabolic clearances following the administration of a 250 tablet to this subject. What is the metabolic clearance?

Part 6 Determine t_{max} and C_{max} for this patient with renal impairment (elimination half-life 14 hours) following a 250 mg oral dose. What is t_{max}?

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